

A study of EEG motor imagery classification algorithm based on WPD-AFBCSP

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Abstract—To improve the accuracy and reliability of EEG motor imagery classification, a WPD-AFBCSP-based EEG motor imagery classification algorithm is proposed. The algorithm first uses the wavelet packet decomposition (WPD) optimised adaptive filter bank co-space pattern algorithm (WPD-AFBCSP) to obtain multidimensional spatial features of the EEG signal, and then the optimal spatial features are obtained by feature selection with the mutual information (MI) feature selection algorithm. Finally, the feature matrix is fed into a deep neural network based on stacked auto-encoders (SAE) for left- and right-handed biclassification to achieve recognition of EEG motor imagery. The method achieves 89.3% classification accuracy on the BCI Competition II Data set III dataset, which can identify EEG motor imagery more effectively and provides a new method for the study of EEG motor imagery.

Keywords—EEG; motor imagery; FBCSP; DNN; SAE

I. INTRODUCTION

As the metaverse concept explodes in popularity, research into the technology associated with it is accelerating. Meta-universes have requirements such as immersion and low latency, which can be met by brain-computer interface motor imagery. Motor imagery (MI) is the most common brain-computer interface paradigm. When people perform unilateral limb activities (e.g. left-handed or right-handed movements), the amplitude of the μ - and β -rhythms in the contralateral primary sensorimotor cortex decreases significantly, while the amplitude of the μ - and β -rhythms in the ipsilateral primary sensorimotor cortex increases significantly, a phenomenon known as event-related synchronization potentials (ERS) and event-related desynchronization potentials (ERD). And motor imagery produces measurable potential changes in the ERS and ERD of the EEG signal. The spatial pattern of ERD during imagined movement of different limbs corresponds to the somatic distribution of motor areas in the cerebral cortex, e.g. when imagining left and right hand movements, distinct ERD

phenomena appear in the region of the contralateral cerebral cortex for the hand, and by identifying ERD and ERS in EEG, the identification of a person's motor imagery can be achieved.

The theory of motion imagery has received a lot of attention from researchers in recent years, and Kavitha P. Thomas et al. proposed discriminative filter bank common spatial pattern (DFBCSP), which reduced the BER by 8.9% compared to FBCSP[1]. Hema C.R., National University of Malaysia, studied the application of feed-forward neural network (FNN) to motor imagery and used dynamic cascade feed-forward neural network (DCFNN) to classify the four states of EEG signals collected by BCI during motor imagery. The DCFNN was used to classify four states of EEG signals collected by BCI during motor imagery[2].

Based on the above research, this paper proposes an optimised adaptive filter bank co-space pattern algorithm based on wavelet packet decomposition, and uses a deep neural network based on stacked self-encoders for classification on this basis.

II. SOURCE OF DATA SET

The BCI Competition II Data set III dataset is an EEG motor imagery dataset created by Alois Schlögl et al. at the Graz University of Technology. This data set records the EEG signals of a normal subject (female, 25 years old). The subject sits in a relaxed chair with armrests. Movements of the left or right hand were imagined by the computer sending random left and right hand cues in advance.

The dataset has 3 channels, C3, Cz and C4. The EEG sampling frequency is 128Hz, filtered between 0.5 ~ 30Hz, and the classification labels are 1 and 2, corresponding to left and right handed respectively. The dataset contains four arrays: x_{train} , y_{train} , x_{test} and y_{test} . Of these, x_{train} is the training set data and is a three-dimensional array representing

140 experiments, each of which acquired three channels of physiological signals, each channel sampled at 128Hz for 9s, for a total of 1152 points (9s× 128Hz). y_{train} is the training set label, a two-dimensional array indicating whether the subject imagined the left or right hand for this experiment.

III. RESEARCH METHODOLOGY

A. Common Spatial Pattern Algorithm (CSP)

The Common Spatial Pattern (CSP) algorithm is a space domain filtering feature extraction algorithm for two classification tasks. The basic principle of the CSP algorithm is to find an optimal set of spatial filters for projection using diagonalisation of matrices to maximise the variance of one class while minimising the variance of the other, taking the approach of diagonalising the covariance matrices of the two classes of tasks simultaneously to obtain the most differentiable feature vector. It is suitable for feature extraction for binary classification tasks. However, the performance of the covariance pattern algorithm depends on whether its operating band is correctly selected, and its classification accuracy is poor if the EEG signal is not filtered or is filtered using an inappropriate band range, or if only the CSP algorithm is used for feature extraction of the EEG signal.

B. Filter Bank Common Space Pattern Algorithm(FBCSP)

The principle of the Filter Bank Common Space Pattern (FBCSP) algorithm is to use a filter bank to decompose the EEG signal into 9 bands from a bandwidth of 4Hz ~ 40Hz, resulting in a feature vector with a higher degree of discrimination[3]. The schematic is shown in Figure 1.

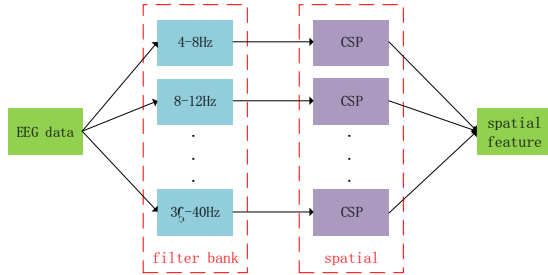


Figure 1. Schematic diagram of the FBCSP.

The FBCSP is divided into two parts.

- Band filtering: Using a Chebyshev II filter to form a filter bank, the bandpass of the EEG signal is divided into 9 bands: 4~8Hz, 8~12Hz, 12~16Hz, 16~20Hz, 20~24Hz, 24~28Hz, 28~32Hz, 32~36Hz, 36~40Hz..
- Spatial filtering: Spatial filtering of each sub-band using a common spatial pattern algorithm to obtain the space domain characteristics of each sub-band..

The Space-domain features obtained by FBCSP have more frequency domain information than the air-domain features of CSP, and also avoid the missing feature information caused by not filtering or filtering the EEG signal using an inappropriate band range[4]. However, the FPCSP divides the frequency

band from 4 to 40 Hz into nine sub-bands, resulting in a high computational cost for the algorithm.

C. Optimized Adaptive Filter Bank Common Space Pattern Algorithm Based on Wavelet Packet Decomposition(WPD-AFBCSP)

To address the drawback of the filter bank co-space pattern (FBCSP) algorithm due to the large computational cost brought about by more band divisions, this paper proposes an optimised adaptive filter bank co-space pattern (WPD-AFBCSP) algorithm based on wavelet packet decomposition.

Wavelet packet decomposition (WPD) is an extension of the wavelet transform. The WPD is performed by filtering the input waveform through a high-pass filter and a low-pass filter to obtain the approximate and detailed components, and finally converting the original signal into 2^j (j is the number of decomposition layers) frequency bands, and constructing the wavelet packet through the dual-scale equation, which can be written as

$$\begin{cases} \psi_{j,n}^{2i}(t) = \sum_k h_k \psi_{j-1,2n-k}^i(t) \\ \psi_{j,n}^{2i+1}(t) = \sum_k g_k \psi_{j-1,2n-k}^i(t) \end{cases} \quad (1)$$

where $\Psi(t)$ is the mother wavelet, j is the number of decomposition layers, n is the node number in that layer, i is the node number, k is the displacement factor, and h_k and g_k are a pair of complementary quadrature mirror filters (QMF). Then the wavelet packet coefficients of the k th band of the j th layer can be calculated by equation (2)."

$$\begin{cases} d_j^{2i}(n) = \sum_k h_k d_{j-1}^i(2n-k) \\ d_j^{2i+1}(n) = \sum_k g_k d_{j-1}^i(2n-k) \end{cases} \quad (2)$$

Take the C3 channel in trial 10 of the training set x_{train} as an example, with a sampling frequency of 128Hz and 384 sampling points (the motion imagery lasts for 3 seconds) The figure below shows the wavelet tree for a 4-layer wavelet decomposition using dmey wavelets

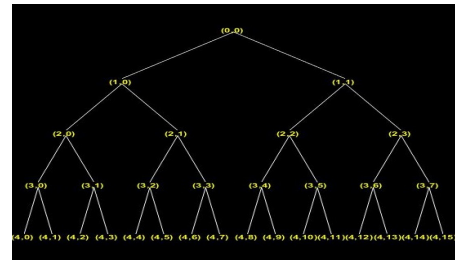


Figure 2. Wavelet packet tree with four layers of wavelet packet decomposition.

The above is the wavelet packet tree, where the nodes are named according to a rule starting from (1, 0), called number 1, (1, 1) is number 2 ... and so on, (4, 0) is number 15, (4, 15) is

number 30. Each node has a corresponding wavelet packet coefficient which determines the magnitude of the frequency. The following figure shows the time-frequency diagram of the wavelet packet decomposition.

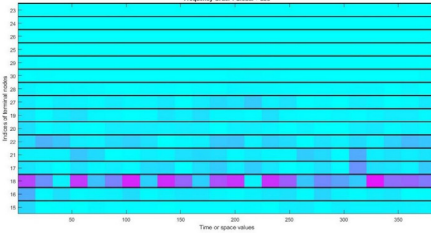


Figure 3. Time-frequency diagram of wavelet packet decomposition.

The X-axis of the time-frequency diagram has 384 points, corresponding to 3 seconds, and each point represents 1/128th of a second. 16=4Hz. the following figure represents the proportion of energy in each band.

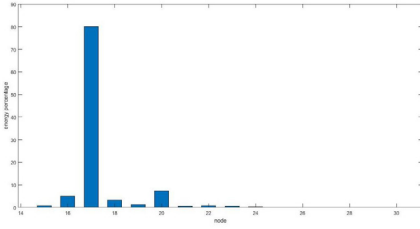


Figure 4. Proportion of energy in each frequency band.

Based on the proportion of energy in each band, the three bands containing the highest proportion of energy are selected, and the FBCSP processing of the nine sub-bands is replaced by the processing of the three sub-bands with the highest proportion of energy. The following diagram shows the schematic of the WPD-AFBCSP algorithm.

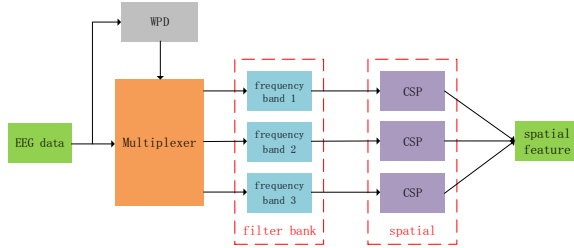


Figure 5. Schematic diagram of the WPD-AFBCSP algorithm.

The WPD-AFBCSP is characterised by adaptively selecting the sub-band of the current EEG signal containing more information as the bandpass band of the filter bank through the WPD algorithm, which avoids the large computational cost of FBCSP and allows the information of the EEG signal not to be missing, so that the features extracted through WPD-AFBCSP are not inferior to The features extracted through the FBCSP are not inferior to those extracted through the FBCSP.

D. Deep Neural Networks based on Stacked Auto Encoders

A self-encoder (AE) is a neural network that is trained to replicate its input at its output. Self-encoders can be used as a tool for learning deep neural networks. Training a self-encoder is unsupervised in the sense that it does not require labelled data. The training process is still based on the optimisation of the cost function. The cost function calculates the reconstruction error of the input x and its output x' .

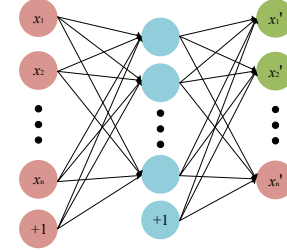


Figure 6. Schematic diagram of the self-encoder

In practice a deep self-encoder is often trained using a layer-by-layer stacking approach, called Stacked Auto-Encoder (SAE)[5]. Stacked Auto-Encoders can generally be trained using Layer-Wise Training to learn the network parameters. In this paper, we increase the number of hidden layers from 1 to 2, which is actually a stack of 2 normal self-encoders. A deep neural network based on stacked self-encoders is shown in the following diagram. The training process is as follows.

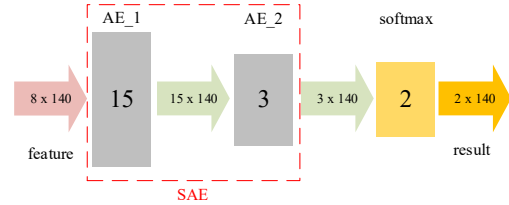


Figure 7. Deep neural network based on stacked self-encoders

First self-encoder: $8 \rightarrow 15 \rightarrow 8$. After training, fix the parameters and the result of the intermediate hidden layer, remove the output layer and the corresponding weight bias, then use the hidden layer 15 as input to the second self-encoder.

Second self-encoder: $15 \rightarrow 3 \rightarrow 15$. After training, fix the parameters and the result of the intermediate hidden layer, remove the output layer and the corresponding weight bias.

A classifier (softmax) is added after the third self-encoder hidden layer for decoding recovery.

IV. ANALYSIS OF EXPERIMENTAL RESULTS

The experimental dataset in this paper is a pre-processed BCI Competition II Data set III dataset, with training data and test data each accounting for 50% of the dataset data. In this paper, the WPD-AFBCSP algorithm was used to extract the EEG signal in the null domain, the MI algorithm was used for feature selection, and a deep neural network formed by stacked self-encoders was used for feature classification to obtain the motion imagery binary classification results. The system

framework is shown in Figure 8. The hardware environment used for the experiments in this paper is a Lenovo ThinkPad E560 laptop and the software environment is matlab 2020b under Windows.

Firstly, we input the null domain features obtained by WPD-AFBCSP into the DNN network after MI feature selection for 30 iterations. The accuracy and loss function images of the model are shown in Figure 9, and the confusion matrix is shown in Figure 10.

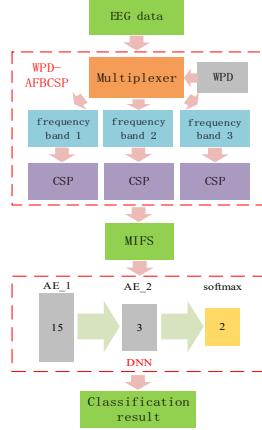


Figure 8. System framework diagram

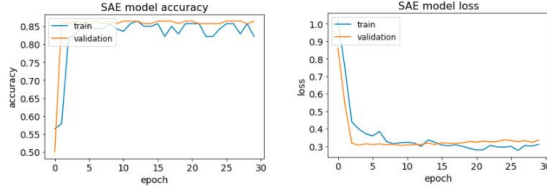


Figure 9. (a) Accuracy of the model after several iterations (b) Image of the cross-entropy loss function of the model

Test Confusion Matrix		
Output Class	1	2
	97 40.7%	2 1.4%
Target Class	13 9.3%	68 48.6%
	81.4% 18.6%	97.1% 2.9%

Figure 10. Confusion matrix for the model

The experimental results show that the algorithm can effectively perform the recognition and classification of motion imagery, and its classification accuracy reaches 89.3%. Meanwhile, in order to explore the influence of the brain number features extracted by WPD-AFBCSP on the classification results, three feature extraction algorithms, WPD, CSP and FBCSP, were selected for experimental comparison with WPD-AFBCSP, and their classification accuracies are shown below.

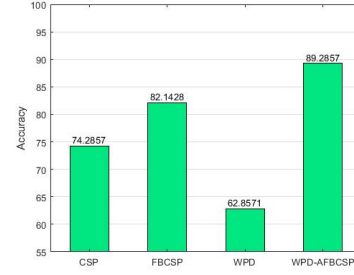


Figure 11. Classification accuracy of multiple feature extraction algorithms

As can be seen from the figure, the correct rate was only 62.8571% when using only WPD for feature extraction, while the accuracy reached 74.2857% when using CSP for feature extraction, due to the high correlation between the null domain features and the EEG motor imagery, which can effectively extract the motor imagery related features. When using FBCSP for feature extraction, the accuracy was improved by 7.8571% compared to CSP, demonstrating the effectiveness of CSP using filter banks to sub-band the raw EEG signal, but this also at the expense of computing speed, with FBCSP taking approximately 9 times longer than CSP. When using WPD-AFBCSP for feature extraction, the accuracy was improved by 7.1429% to 89.2857% compared to that of FBCSP, due to the use of WPD to select the optimal 3 bands, which not only made it possible to reduce the number of filters in the filter bank from 9 in FBCSP to 3, saving 6 times of CSP for each trial. The results not only reduce the number of filters in the filter set from 9 to 3, saving 6 CSPs per trial and improving the computational speed, but also remove the redundant bands, making the extracted features more distinguishable. The results show that the WPD-AFBCSP-based EEG motor imagery feature extraction algorithm has a high accuracy rate and can effectively achieve the feature extraction of EEG motor imagery.

V. Conclusions

In this paper, an optimised adaptive filter bank co-space pattern algorithm (WPD-AFBCSP) based on wavelet packet decomposition is proposed, on which a deep neural network based on stacked self-encoders is used for classification. The wavelet packet decomposition is used to extract band energy features, thus enabling the optimisation of the filter bank common-space pattern algorithm to achieve adaptive selection of the optimal band based on band energy features for air domain feature extraction. Compared with other methods such as WPD, CSP and FBCSP, the accuracy of motion imagery feature recognition after DNN classification is significantly improved. The experimental results demonstrate the feasibility and effectiveness of the method, and provide a new idea for EEG motion imagery recognition.

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